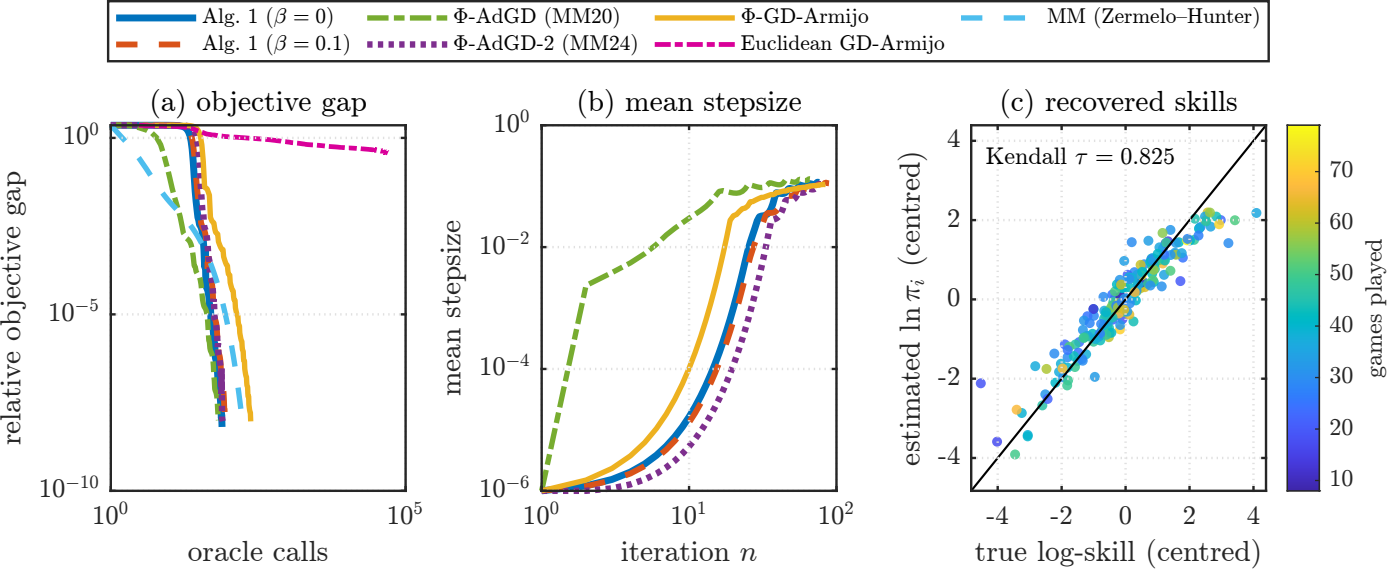

Alg. 1 ($\beta = 0$)
Alg. 1 ($\beta = 0.1$)
Φ-AdGD (MM20)
Φ-AdGD-2 (MM24)
Φ-GD-Armijo
Euclidean GD-Armijo
MM (Zermelo–Hunter)
(a) objective gap
relative objective gap
10^{0}
10^{-5}
10^{-10}
10^{0}
10^{5}
oracle calls
(b) mean stepsize
mean stepsize
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{0}
10^{1}
10^{2}
iteration n
(c) recovered skills
Kendall $\tau = 0.825$
estimated $\ln \pi_i$ (centred)
4
2
0
-2
-4
-4
-2
0
2
4
true log-skill (centred)
games played
70
60
50
40
30
20
10